

5	-90°	-90°	5.71°	-0.986°	-140.55°
10	-90°	-157.38°	26.56°	-0.573°	-174.86°
50	-90°	-168.54°	45°	-2.862°	-223.64°
100	-90°	-177.7°	78.69°	-5.71°	-219.28°
∞	-90°	-178.85°	84.28°	-26.56°	-215.58°
	-90°	-90°	$+90^\circ$	-45°	-229.56°
				-90°	-180°

slm is unstable w.r.t w.r.g

⑥ Sketch the Bode plot for the slm having

$$G(s)H(s) = \frac{4}{(0.1s + 1)^2(0.01s + 1)}$$

find G_m , P_m , ω_{gc} , ω_{pc} & stability.

Soln ⑥ $\rightarrow$

⑦ $G_m = 60\text{ dB}$ $P_m = 25^\circ$

Soln ⑦ $\Rightarrow G(s)H(s) = \frac{K}{s(1+0.9s)(1+0.05s)}$ TCF

⑧ $\Rightarrow K$ is unknown

① here K is unknown so draw the magnitude plot without K . The effect of K is to shift the magnitude plot upwards or downwards by $20\log(K)$. Thus

② Shift the magnitude plot so as to match the given specifications. Equating this shift

to $\log K$, the value of K can be determined.

- ⑤ One simple pole at origin $\frac{1}{s}$ slope = -20 dB/dec
- ⑥ One simple pole $\frac{1}{1+0.2s} \Rightarrow T_1 = 0.2 \text{ sec}, \omega_{c1} = 5\text{ rad/sec}$
- ⑦ One simple pole $\frac{1}{1+0.05s} \Rightarrow T_2 = 0.05 \text{ sec}, \omega_{c2} = 20\text{ rad/sec}$

Terms	Crossover freq (Hz)	Slope	Change in slope
$\frac{1}{s}$	—	-20 dB/dec	-20 dB/dec
$\frac{1}{1+0.2s}$	5 rad/sec	-20 dB/dec	-40 dB/dec
$\frac{1}{1+0.05s}$	20 rad/sec	-20 dB/dec	-60 dB/dec

$\Rightarrow$ ④ Phase plot $\Rightarrow \frac{1}{s} = \frac{1}{j\omega} = -\tan^{-1}(\infty) = -90^\circ$

$\rightarrow \frac{1}{1+0.2s} = \frac{1}{1+0.2j\omega} = -\tan^{-1}(0.2\omega),$

$\rightarrow \frac{1}{1+0.05s} = -\tan^{-1}(0.05\omega)$

ω rad/sec	0.5	5	10	20	50	∞
ω	-90°	$-\tan^{-1}(0.2\omega)$	$-\tan^{-1}(0.05\omega)$	ϕ_p		
0.5	-90°	-5.71°	-1.43°	-97.14°		
5	-90°	-45°	-14.04°	-149.03°		
10	-90°	-63.43°	-26.56°	-180°		
20	-90°	-75.96°	-45°	-210.96°		
50	-90°	-84.29°	-63.19°	-242.5°		
∞	-90°	-90°	-90°	-270°		

Steps to find 'K' for the given margin

- > Draw the horizontal line below 0 db at a distance of given gain margin till it intersects the vertical line of wpc. Mark this point as A'
- > Draw the vertical line from wpc till it intersects the magnitude plot without K. Mark this point as A.
- > The point A must be at A' to match the given gain margin, hence A to A' is the shift required.
- > Upward shift must be taken as positive and downward shift as negative.
- > Equate $20 \log K = \text{shift of A to A'}$ shift (A A')

$$\Rightarrow 20 \log K = 22.08$$

$$\log K = 1.1 \quad (01), \quad \log K = 1$$

$$K = \frac{12.6}{\boxed{K=10}}$$

Steps to find K for given phase margin (Pm)

- ① Draw the horizontal line above -180° line at a distance of given phase margin till it intersects the phase plot. Mark this point as 'B'. draw the vertical line from B line it intersects the magnitude plot without K. Mark this point as C'.

② This C' must be on 0dB line at ' C ' as the vertical line through point B must be wgc line for the given phase margin.

③ Thus distance of C' from 0dB line is the shift required to satisfy given phase margin.

④ Upward shift must be taken as positive & downward shift as negative.

$$\Delta \log K = \text{shift} + (C'C)$$

$$\Delta \log K = 18 \text{ dB}$$

$$K = 7.9 \approx \underline{\underline{8}}$$

⑤ The Open loop TF $G(s)H(s) = K \frac{S(1+s)(1+0.1s)}{S(1+s)(1+0.1s)}$

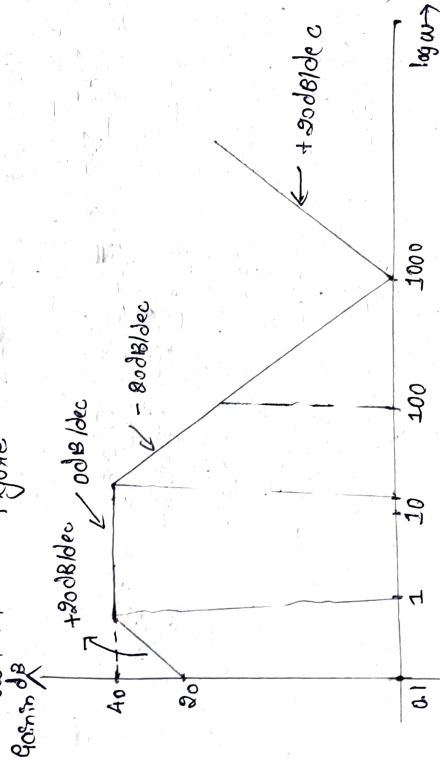
determine the values of K such that (a) $G_m = 10 \text{ dB}$

(b) $PM = 24^\circ$ Use Bode plot to find K

$$\text{Soln (a) Ans } K = 3.16,$$

Type III

⑥ Find the TF of the s/m whose Bode diagram is shown in the figure



Soln Note :- $-20\text{dB} \rightarrow \text{pole} \quad ; \quad +20\text{dB} \rightarrow \text{zero}$

① To find K :- Always consider $\omega = 1$

here $\log K = 40\text{dB}$

$$[K = 100]$$

② (a) Initial slope is $+100\text{dB/dec}$. $\therefore$ There is one zero

at the origin $\rightarrow \text{factor} = s$

(b) At $\omega = 1$, the change in slope, from $+20\text{dB}$ to 0dB
 $+20\text{dB} - 0\text{dB} = 20\text{dB}$ p.e. By ^{adding} -20dB/dec hence,

a simple pole with $\omega_{c1} = 1 \quad \therefore T_1 = \frac{1}{\omega_{c1}} = 1$

Hence, the factor is $\frac{1}{1+s} = \frac{1}{1+s}$

(c) At $\omega = 10$, the change in slope is from 0dB to -20dB/dec

i.e. By adding -20dB/dec hence a simple pole exists
 with $\omega_{c2} = 10\text{ rad/sec} \quad \therefore T_2 = \frac{1}{\omega_{c2}} = 0.1$ with

$$\text{factor} = \frac{1}{1+0.1s}$$

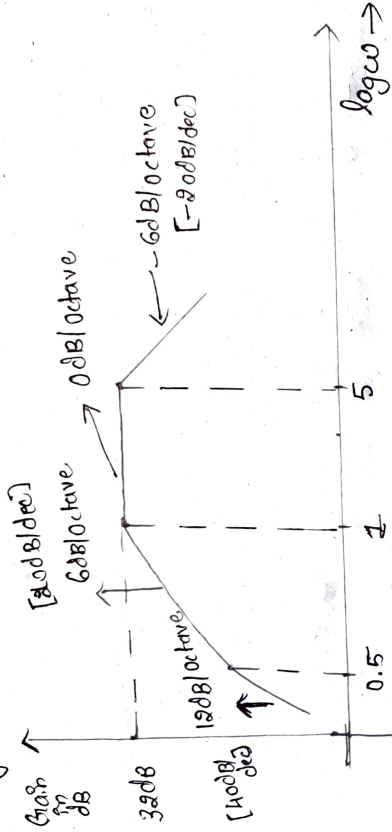
(d) At $\omega = 1000$, the change in slope from -20dB/dec to $+20\text{dB/dec}$ is $-20\text{dB} + 40\text{dB} = +20\text{dB}$ adding $+20\text{dB/dec}$

hence a simple zero exists with $\omega_{c3} = 1000$, $T_3 = 0.001$
 of second order

$$\text{factor} = \frac{1}{(1+0.001s)^2} \Rightarrow (1+0.001s)^2$$

$$\therefore \text{The TF is} = \frac{100s(1+0.001s)}{s(1+s)(1+0.1s)} = \frac{K(zeros)}{poles}$$

10) Find the TF for the Bode plot shown in the figure.



Soln Note :- 6 dB/octave = 20 dB/dec
12 dB/octave = 40 dB/dec

1) $20 \log k = 80 \text{ dB}$

$k = 39.81 \approx 40$

2) At $\omega = 0.5 \text{ rad/sec}$, slope change slope of 120 dB/dec initial slope, a zero exist at origin with factor (s)

At $\omega = 0.5 \text{ rad/sec}$, change 40 dB to 80 dB $\therefore 0 \text{ dB} \rightarrow 80 \text{ dB}$

$\rightarrow$ a simple pole $\rightarrow \omega c_1 = 0.5 \quad T_1 = 2 \Rightarrow \frac{1}{1+s}$

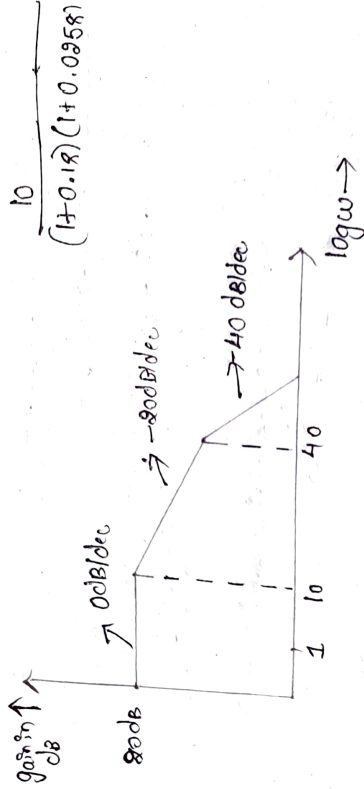
c) At $\omega = 1$, slope change from 80 dB/dec to 0 dB add -20 dB/dec. $\omega c_2 = 1, \quad T_2 = 1 \Rightarrow \frac{1}{1+s}$ (factor)

d) $\omega = 5$:- slope change from 0 to -20 dB add -20 dB. $\rightarrow \omega c_3 = 5 \quad T_3 = \frac{1}{5} = 0.2 \Rightarrow \frac{1}{1+0.2s}$

TF = $\frac{40s^2}{(1+s)(1+s)(1+0.2s)} = \frac{40s^2}{(s+1)(s+1)(0.2s+1)}$

(11) Determine TF of s/m whose Bode diagram is

shown below.



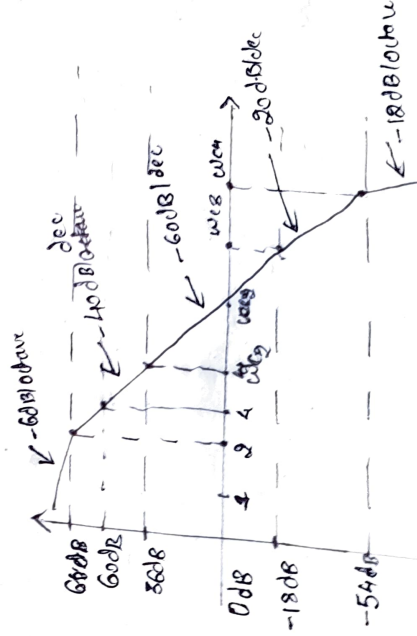
Soln (a) For K at $\omega_c = 1$, $20 \log K = 80 \text{ dB}$

$$20 \log K = 80 \Rightarrow K = 10$$

(b) (i) At $\omega_c = 10$, change in slope $\rightarrow 0$ to -20 dB

add -20 dB $\therefore$ First order $T_1 = 0.1 \Rightarrow \frac{1}{1+0.1s}$

(12) For the open loop TF of a unity f/B s/m whose Bode plot is shown below. find the TF



Soln > To find K, find the Equation of line

of slope -40dB/dec $y = -40\log(\omega) + C_1$

at $\omega = 4$, $y = 60\text{dB} \Rightarrow 60\text{dB} = -40\log(4) + C_1$

$\Rightarrow 60\text{dB}$
 $60 + 40\log(4) = C_1$

$C_1 = 84$

$y = -40\log(\omega) + 84$

$\therefore$ At $\omega = 2$, $y = ? \Rightarrow y = -40\log(2) + 84$

$y = 72\text{dB}$

Now $\therefore$ The Eqⁿ of straight line with slope -20dB/dec

$\therefore y = -20\log(\omega) + C_2$

> At $\omega = 2$, $y = 72 \Rightarrow 72 + 20\log(2) = C_2$

$C_2 = 78$

> $y = -20\log(\omega) + 78 \rightarrow$ at $\omega = 1$

$y = 78\text{dB}$

$\Rightarrow 20\log K = 78 \quad \therefore K = 7943$

(a) Initially slope $= -20\text{dB/dec} \Rightarrow$ Pole at origin $\Rightarrow \frac{1}{s}$ factor

(b) Slope change -20dB to $-40\text{dB} \rightarrow$ at $\omega_1 = 2$

$\therefore$ add $-20\text{dB} \Rightarrow \omega_1 T_1 = 0.5 \rightarrow \text{Pole at } \frac{1}{1+0.5s}$

(c) Slope change $\therefore -40\text{dB}$ to -60dB at $\omega_2 = ?$

To find ω_2 , $y = 30\text{dB}$. $\therefore y = -60\log(\omega) + C$

using $y = -40\log(\omega_2) + C$, $C_1 = 84$

$36 - 84 = -40\log(\omega_2)$

$\frac{-48}{-40} = \log(\omega_2) \Rightarrow$

$1.2 = \log(\omega_2) \Rightarrow \omega_2 = 16\text{rad/sec}$

$T_2 = 0.06$. $\therefore \frac{1}{1+0.06s}$

(d) ω_3 find $\rightarrow$ wk at $\omega_3 = 16$, $y = 36\text{dB}$

$y = -40\log(\omega_3) + C_3$

$36 + 40\log(16) = C_3$

$C_3 = 108$

$$y = -60 \log(16) + T_3 \times$$

$$C_3 = 108$$

$$y = -18 \text{ dB}$$

$$y = -60 \log(16) + 108$$

$$y = -60 \log(\omega_{c3}) + C_3$$

$$y = -18 \text{ dB}$$

$$-18 + 60 \log(16)$$

$$-18 - 108 = -60 \log(\omega_{c3})$$

$$\boxed{\omega_{c3} = 126 \text{ rad/sec}}$$

$$T_3 = 0.008$$

$\Rightarrow$ Change from -460 to -80 change of $+40 \text{ dB}$.

$\therefore$ Zero exists of second order $\Rightarrow \frac{1}{(1 + 0.008s)^2}$

e) $\omega_{c4} \therefore y = -20 \log(\omega_{c4}) + C_4$

$\Rightarrow$ To find $C_4 \Rightarrow y = -20 \log(\omega_{c3}) + C_4$

$$-18 = -20 \log(126) + C_4$$

$$\boxed{C_4 = 24}$$

Now $y = -54 \text{ dB}$

$$y = -20 \log(\omega_{c4}) + C_4$$

$$-54 = -20 \log(\omega_{c4}) + 24$$

$$\omega_{c4} = 7943 \text{ rad/sec}$$

$$T_4 = 0.0001$$

change from -20 dB to $-40 \text{ dB} \Rightarrow -20 \text{ dB/dec}$

$\therefore$ pole exist $\Rightarrow \frac{1}{(1 + 0.0001s)}$

$$\therefore TF = \frac{7943 (1 + 0.00793s)^2}{s[(1 + 0.5s)(1 + 0.06s)(1 + 0.008s)^2(1 + 0.0001s)]}$$